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Batch: 15

Experiment-46

Algorithm

Main Algorithm

1. Start.
2. Read the size of the array size.
3. Declare a variable-length integer array nums[size].
4. Read size elements into the array.
5. Pass the address of nums and size to the function Bubble_sort().
6. In Bubble_sort():
    - Compare adjacent elements.
    - If the first element is greater than the next element, swap them.
    - Repeat the process until the entire array is sorted in ascending order.
7. Display the sorted array.
8. Pass the address of the sorted array and size to the function occurrence().
9. In occurrence():
    - Read the integer target.
    - Initialize first = -1 and last = -1.
    - Traverse the sorted array using a pointer.
    - Whenever target is found:
        - If first == -1, store the current index in first.
        - Store the current index in last.
10. If first == -1, display -1, -1 for the first and last occurrences.
11. Otherwise, display the **value and index** of the first occurrence.
12. Display the **value and index** of the last occurrence.
13. Stop.

Test Case Table

Test Case	Size	Input Array	Target	Sorted Array	First Occurrence (Value, Index)	Last Occurrence (Value, Index)
1	8	5 2 7 2 9 2 4 6	2	2 2 2 4 5 6 7 9	(2, 0)	(2, 2)
2	6	4 1 3 4 2 4	4	1 2 3 4 4 4	(4, 3)	(4, 5)
3	5	10 20 30 40 50	30	10 20 30 40 50	(30, 2)	(30, 2)
4	5	5 1 3 2 4	8	1 2 3 4 5	(-1, -1)	(-1, -1)
5	5	7 7 7 7 7	7	7 7 7 7 7	(7, 0)	(7, 4)

C Program

```
#include <stdio.h>

void Bubble_sort(int *nums, int size);
void occurrence(int *nums, int size);

int main()
{
    int size;

    printf("Enter the size of array: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter %d elements:\n", size);
    for (int i = 0; i < size; i++)
    {
        scanf("%d", &nums[i]);
    }

    Bubble_sort(nums, size);

    printf("Sorted array: ");
    for (int i = 0; i < size; i++)
    {
        printf("%d ", nums[i]);
    }
```

```
    printf("\n");

    occurrence(nums, size);

    return 0;
}

void Bubble_sort(int *nums, int size)
{
    int temp;

    for (int i = 0; i < size - 1; i++)
    {
        for (int j = 0; j < size - i - 1; j++)
        {
            if (*(nums + j) > *(nums + j + 1))
            {
                temp = *(nums + j);
                *(nums + j) = *(nums + j + 1);
                *(nums + j + 1) = temp;
            }
        }
    }
}

void occurrence(int *nums, int size)
{
    int target;
    int first = -1, last = -1;
```

```
    printf("Enter target element: ");
    scanf("%d", &target);

    for (int i = 0; i < size; i++)
    {
        if (*(nums + i) == target)
        {
            if (first == -1)
            {
                first = i;
            }

            last = i;
        }
    }

    if (first == -1)
    {
        printf("First occurrence: -1, -1\n");
        printf("Last occurrence: -1, -1\n");
    }
    else
    {
        printf("First occurrence: Value = %d, Index = %d\n",
               *(nums + first), first);

        printf("Last occurrence: Value = %d, Index = %d\n",
               *(nums + last), last);
    }
}
```

Compilation And Execution